
AutoInterpBench: Level-1 Grounded Benchmarking for Autonomous Mechanistic Interpretability Discovery

Abstract

Mechanistic interpretability (MI) has produced rich qualitative insights—induction heads, knowledge neurons, linear safety directions—but these remain isolated case studies on individual models. We argue that advancing MI requires not just better methods but *autonomous systems* that can reproduce, diagnose, and discover interpretability phenomena at scale. We introduce **AutoInterpBench**, a three-level benchmark that evaluates MI agents along a hierarchy of scientific competence: **Level 1 (Reproduce)**: execute a specified GPU experiment and verify that results match ground truth within numerical tolerance (200 tasks); **Level 2 (Diagnose)**: given an observed cross-model anomaly, select appropriate methods and identify root causes (34 tasks); **Level 3 (Discover)**: conduct open-ended exploration and produce novel, validated findings, evaluated by LLM-as-Judge (30 tasks). The benchmark is grounded in **764 real GPU experiments** across 14 models (0.6B–8B, including 2 vision-language models), 80 experiment types, and 5 controlled comparison dimensions (architecture generation, scale, modality, thinking mode, instruction tuning). We propose a **harness-driven discovery** framework in which Level 1 benchmark scores serve as fitness signals for refining MI analysis primitives. A zero-shot Claude Sonnet 4.6 baseline achieves only 0.20 on Level 1 and 0.20 on Level 2, confirming that current frontier LLMs lack the domain expertise for autonomous MI research. Through systematic cross-model analysis enabled by the benchmark’s 764 experiments, we identify three empirical regularities of MI—linear representation universality, MLP-dominated factual memory, and instruction-tuning reorganization—that hold across all tested models and multiple independent methods.

1 Introduction

Mechanistic interpretability—the effort to reverse-engineer the algorithms learned by neural networks [Olah et al., 2020, Elhage et al., 2021]—has produced a rich catalog of phenomena: induction heads [Olsson et al., 2022], knowledge-storing MLP rows [Meng et al., 2022], sparse overcomplete features [Cunningham et al., 2023, Bricken et al., 2023], and linear safety directions [Arditi et al., 2024]. Yet these discoveries share a fundamental limitation: they are *isolated case studies* on individual models, relying on researcher intuition for hypothesis generation, experiment design, and result interpretation.

Recent work on AI-driven scientific discovery [Lu et al., 2024, Romera-Paredes et al., 2024, Si et al., 2024] demonstrates that language models can generate novel hypotheses and even write executable experiments. However, two critical gaps remain. First, there is no *benchmark* that evaluates an agent’s ability to do MI research at different levels of scientific competence—from basic reproduction to genuine discovery. Existing MI evaluations are either text-matching QA [Bills et al., 2023] or single-phenomenon case studies [Shaham et al., 2024]. Second, there is no *mechanism* that uses objective experimental feedback to systematically improve an agent’s MI capabilities over time.

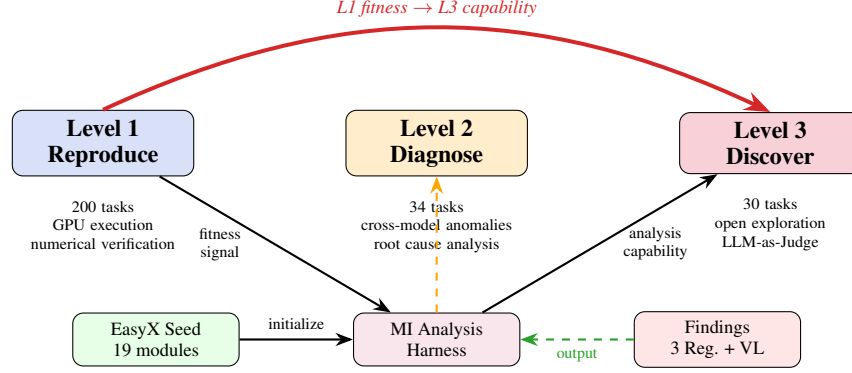


Figure 1: AutoInterpBench framework. Level 1 benchmark scores provide objective fitness signals for refining the MI analysis harness. The harness, seeded with EasyX (19 modules), is expanded through Level 1-guided development, building the analysis capability needed for Level 3 discovery—enabling three empirical regularities and cross-model insights to be identified across 14 models.

We address both gaps by introducing a three-level framework for autonomous MI research:

1. **Level 1: Reproduce** (*know that*). Given a specified experiment and model, execute the GPU computation and verify that results match ground truth within numerical tolerance. This tests operational competence: can the agent use MI tools correctly?
2. **Level 2: Diagnose** (*know why*). Given an observed anomaly (e.g., “identity crisis depth shifts from 97% to 17% after instruction tuning”), select appropriate experiments and identify the root cause. This tests analytical reasoning: can the agent design informative experiments?
3. **Level 3: Discover** (*find new*). Given an open research direction, conduct original exploration and produce novel, validated findings. This tests scientific creativity: can the agent generate knowledge that humans have not yet documented?

The key insight is that **Level 1 provides an objective, automatically verifiable signal** (do the numbers match?), while Level 3 requires subjective judgment of novelty and significance. We exploit this asymmetry: Level 1 benchmark scores serve as *fitness signals* for refining MI analysis primitives—a **harness-driven discovery** framework that builds Level 3 discovery capability from Level 1 supervision. This is inspired by how AutoHarness [Lou et al., 2026] uses action-legality feedback to synthesize code harnesses that constrain LLM agent actions.

Our contributions are:

1. **AutoInterpBench**: a three-level benchmark (200 + 34 + 30 = 264 tasks) grounded in 764 real GPU experiments across 14 models, 80 experiment types, and 5 comparison dimensions (§2).
2. **Harness-driven discovery framework and AutoDiscover v2**: a framework in which Level 1 scores guide the refinement of MI analysis primitives, instantiated as a four-agent system. Through systematic exploration, we identify three cross-model empirical regularities of MI (§3).
3. **Five-dimension cross-model analysis**: controlled comparisons across architecture generation, scale, modality, thinking mode, and instruction tuning on 14 models including 2 vision-language models (§4).

2 AutoInterpBench

2.1 Three Levels of Scientific Competence

We formalize three levels of scientific competence for MI agents, inspired by the epistemological distinction between phenomenological knowledge (*know that*), mechanistic understanding (*know why*), and axiomatic discovery (*find new*).

Table 1: Three levels of MI scientific competence with evaluation methodology.

Level	Input	Output	Evaluation	Analogy
L1: Reproduce	experiment spec	numerical results	tolerance match	lab technician
L2: Diagnose	observed anomaly	root cause + evidence	rubric scoring	diagnostician
L3: Discover	research direction	novel findings	LLM-as-Judge	scientist

Level 1: Reproduce. The agent receives a fully specified experiment: model name, experiment type, and expected metric ranges. It must load the model, execute the experiment on a GPU, and return numerical results. Evaluation is *objective*: each metric is compared against ground truth with a tolerance (typically $\pm 10\%$ for floats, ± 2 for layer indices, exact for booleans). The score is the fraction of metrics that pass. This level tests whether the agent can operate MI tools correctly—analogueous to a lab technician following a protocol.

Level 2: Diagnose. The agent receives a description of an observed phenomenon—typically a cross-model difference discovered in our five-dimension analysis. For example: “Qwen3-4B-Base exhibits its identity crisis at 97.2% depth, while Qwen3-4B (instruct) shows it at 17.4%. This pattern holds across all base/instruct pairs.” The agent must hypothesize a cause, propose experiments to test it, and interpret results. Evaluation uses a four-dimension rubric: method selection (30%), diagnosis accuracy (30%), evidence quality (20%), and explanation clarity (20%). This tests analytical reasoning—analogueous to a diagnostician interpreting symptoms.

Level 3: Discover. The agent receives a broad research direction (e.g., “Investigate how vision-language models differ from text-only models in internal representations”) and must produce original findings. Evaluation is by *LLM-as-Judge* scoring four dimensions: novelty (25%), validity (25%), significance (25%), and methodology (25%). This tests scientific creativity—analogueous to a researcher conducting original investigation.

The three levels are deliberately ordered by objectivity of evaluation: Level 1 is fully automatic, Level 2 requires structured rubric evaluation, and Level 3 requires subjective assessment. This ordering is critical for the harness-driven discovery framework (§3).

2.2 Ground Truth and Benchmark Construction

AutoInterpBench is a 264-task benchmark grounded in 764 real GPU experiments. Unlike prior MI evaluations that rely on text matching [Bills et al., 2023] or single-model case studies [Shaham et al., 2024], our benchmark requires agents to *actually execute experiments on GPUs* and produce *numerically verifiable results*.

The benchmark’s ground truth is derived from a systematic experimental campaign: **80 experiment types** organized into 15 mechanistic families (attention circuits, sparse features, representation geometry, layer roles, causal steering, safety directions, in-context learning, hallucination detection, feedforward memory, cross-lingual transfer, emergent capabilities, long context, medical hallucination, financial hallucination, features); **14 models** spanning 4 architecture families—Qwen3 (0.6B, 1.7B, 4B, 8B instruct; 1.7B, 4B, 8B base; 4B-Instruct-2507), Qwen2.5 (1.5B, 7B instruct), Llama-2-7B, OLMo-2-1B, and 2 vision-language models (Qwen2.5-VL-7B, LLaVA-1.5-7B); **5 comparison dimensions** enabling controlled analysis (Table 2).

All experiments are executed using our **EasyX** toolkit (v2.4), a hook-based analysis framework with 19 modules and unified support for both causal language models and vision-language architectures. The full experimental log (`real_findings.jsonl`) contains 817 runs covering 764 unique (experiment, model) cells with structured result dictionaries.

Construction process. The ground truth was assembled through a systematic three-phase grid search over a grid $\mathcal{A} \times \mathcal{B} \times \mathcal{C}$, where $\mathcal{A} = 80$ MI experiment types (derived from a six-survey gap analysis), $\mathcal{B} = 14$ models, and $\mathcal{C} = 5$ comparison dimensions. Phase 1 ran 60 core experiments on 12 text models (720 cells); Phase 2 added 20 survey-identified experiment types on a primary analysis model (20 cells); Phase 3 extended to 2 VL models (24 cells)—yielding $720 + 20 + 24 = 764$ unique cells. Benchmark tasks were then extracted from this grid at three difficulty levels. The full construction methodology, including the survey audit and gap analysis, is detailed in Appendix A.

Table 2: Five controlled comparison dimensions. Each dimension isolates one variable while controlling others.

Dimension	Comparison Groups	Controlled Variable
Architecture generation	Qwen3-4B vs Qwen2.5-7B vs LLaMA-2-7B	similar scale, different generation
Scale	Qwen3 0.6B vs 4B vs 8B	same architecture, different size
Modality	Qwen2.5-7B vs Qwen2.5-VL-7B vs LLaVA-1.5-7B	same scale, text vs VL
Thinking mode	Qwen3-4B (thinking) vs Qwen3-4B-2507 (no thinking)	same weights, different template
Instruction tuning	Qwen3-4B-Base vs Qwen3-4B	same architecture, base vs instruct

2.3 Level 1 Tasks: Reproduce

We generate 200 Level 1 tasks by sampling from the 764 completed experiments, ensuring coverage of all 14 models and diverse experiment types. Each task specifies an experiment name, model name, and a list of required metrics. Ground truth is extracted directly from the experimental results, with tolerances: $\pm 10\%$ relative for floats, ± 2 absolute for layer indices, exact match for booleans.

Example task: “Run `residual_stream_norm_growth` on Qwen3-8B. Report: `growth_ratio`, `num_phase_transitions`, `crisis_layer_fraction`.” **Ground truth:** `growth_ratio` = 122.7 $\pm$ 12.3, `num_phase_transitions` = 2 $\pm$ 2, `crisis_layer_fraction` = 0.188 $\pm$ 0.019.

2.4 Level 2 Tasks: Diagnose

We generate 34 Level 2 tasks from notable cross-model differences identified in the five-dimension analysis. Each task presents an observed phenomenon with specific numbers drawn from real experiments, and asks the agent to hypothesize, design verification experiments, and explain.

Example task: “OBSERVATION: Qwen3-4B (instruct) shows effective steering vectors (`steering_effective=True`, `norm=16.95`), while Qwen2.5-7B shows ineffective steering (`norm=22.73`). LLaMA-2-7B also fails (`norm=17.42`). TASK: (1) Hypothesize why only Qwen3 instruct enables steering. (2) Design experiments to test your hypothesis. (3) Predict expected results.”

Tasks span all five comparison dimensions: scale effects (13 tasks), base vs instruct differences (9), architecture generation (9), modality differences (2), and thinking mode (1). The uneven distribution reflects the underlying data: scale and instruction-tuning comparisons are supported by 6+ model pairs each, while thinking mode is a single controlled pair (Qwen3-4B with/without thinking template) and modality is limited to 2 VL models with only 12 experiments each. New: We generate Level 2 tasks through a reverse-engineering process applied to structured text mining of mechanistic interpretability literature. Each task presents an isolated, microscopic anomaly (OBSERVATION) derived from real experimental triggers, deliberately withholding the root cause and specific intervention methods. The agent is required to output a complete diagnostic plan.

Example task: “OBSERVATION: When performing linear concept erasure, correlation probes indicate the information is ‘completely erased’ (low accuracy), but downstream tasks can still extract the concept, revealing severe statistical leakage. TASK: Provide a complete diagnostic plan, including candidate hypotheses, experimental design, and how you expect to update your conclusions based on the evidence.”

Evaluation shifts from static metric weighting to an LLM-as-Judge, measuring alignment against a 6-step Golden Answer rubric reverse-engineered from the source paper’s experimental trace. Crucially, the agent is evaluated on its ability to formulate mutually exclusive hypotheses—incorporating historically rejected paths (*Rejected_H hypotheses*) *as distractors*—and design verifiable experiments with explicit observable expectations.

2.5 Level 3 Tasks: Discover

We construct 30 Level 3 tasks as open-ended research prompts. Each specifies a research direction, available models, and minimum experiment count, but leaves the hypothesis, method selection, and interpretation entirely to the agent.

Table 3: Comparison with related scientific agent benchmarks.

Benchmark	#Tasks	Domain	GPU exec	Multi-level	Self-expand	Best score
DiscoveryBench [Majumder et al., 2024]	1,167	General	×	×	×	25%
ScienceAgentBench [Chen et al., 2025]	102	Multi-sci	partial	×	×	34.3%
MLAgentBench [Huang et al., 2024]	13	ML	✓	×	×	37.5%
MAIA [Shaham et al., 2024]	—	Vision MI	✓	×	×	—
AutoInterpBench (ours)	264	LLM MI	✓	✓	✓	0.20 (L1)

Example task: “Investigate how vision-language models (Qwen2.5-VL-7B, LLaVA-1.5-7B) differ from text-only counterparts in internal representations. You have access to 80 experiment types. Run at least 3 experiments and report novel findings.”

Evaluation uses LLM-as-Judge with a structured rubric scoring novelty (25%), validity (25%), significance (25%), and methodology (25%). New: We construct Level 3 tasks to challenge the agent with out-of-domain scenarios, focusing on profound theoretical conflicts or counter-intuitive anomalies where existing paradigms fail. Unlike Level 2 tasks that evaluate the adept application of known tools, Level 3 requires the agent to pioneer novel mechanistic paradigms (e.g., dissecting MoE routing or System 2 thinking processes).

Example task: “OBSERVATION: Standard activation patching fails to localize reasoning processes in dynamic MoE routing, suggesting a distributed mechanism not captured by static, independent layer interventions. TASK: Provide a complete diagnostic plan to establish a new explanatory paradigm, detailing your hypotheses, experimental interventions, and criteria for belief updates.”

Evaluation also employs the 6-step rubric via LLM-as-Judge but heavily penalizes reliance on conventional in-domain explanations. The scoring strongly emphasizes the agent’s experimental trace (Trace)-specifically, how dynamically the agent updates its internal beliefs in response to mixed experimental feedback to discard false hypotheses and isolate the true causal root.

2.6 Scoring

Each level has a distinct scoring mechanism:

Level 1: $S_1 = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \mathbf{1}[|v_m - v_m^*| \leq \tau_m]$, where $\mathcal{M}$ is the set of required metrics, v_m is the agent’s reported value, v_m^* is ground truth, and τ_m is the tolerance.

Level 2: $S_2 = 0.30 \cdot \text{method} + 0.30 \cdot \text{diagnosis} + 0.20 \cdot \text{evidence} + 0.20 \cdot \text{explanation}$, evaluated by structured rubric.

Level 3: $S_3 = 0.25 \cdot \text{novelty} + 0.25 \cdot \text{validity} + 0.25 \cdot \text{significance} + 0.25 \cdot \text{methodology}$, evaluated by LLM-as-Judge.

Aggregate: $S = 0.40 \cdot \bar{S}_1 + 0.30 \cdot \bar{S}_2 + 0.30 \cdot \bar{S}_3$. Level 1 receives the highest weight because it is the most objectively grounded (fully automated, numerically verifiable), while Levels 2 and 3 share the remaining weight equally as they both involve rubric-based or judge-based evaluation with inherent subjectivity.

2.7 Comparison with Existing Benchmarks

AutoInterpBench is unique in combining: (1) mandatory GPU execution (not text matching), (2) a multi-level evaluation hierarchy with an objective-to-subjective gradient, (3) a self-expanding design where Level 3

discoveries can become new Level 1 tasks, and (4) grounding in real MI experiments rather than synthetic or literature-derived tasks.

3 Harness-Driven MI Discovery and AutoDiscover v2

The central hypothesis of this paper is that optimizing for Level 1 reproduction implicitly builds Level 3 discovery capability, because both require the same underlying competencies: correct tool usage, meaningful experiment design, and sound interpretation of results. We exploit the objectivity asymmetry of our three levels: Level 1 provides a fully automatic, numerically grounded fitness signal, while Level 3 discovery capability grows as a consequence of the harness built through Level 1-guided development. This section describes the harness framework (the “what”), its refinement mechanism (the “how”), and its instantiation as the AutoDiscover v2 multi-agent system (the “who”).

3.1 Harness as Structured Analysis Capability

We define an MI *harness* $\mathcal{H} = (\mathcal{P}, \mathcal{V})$ consisting of:

Analysis primitives $\mathcal{P} = \{p_1, \dots, p_K\}$: executable functions that take a model and return structured results. Examples include `run_logit_lens(model, layers)`, `run_activation_patch(model, component)`, and `run_linear_probe(model, concept, layer)`. Our full harness contains $K = 80$ primitives from the EasyX toolkit; the evolution process (§4.3) starts from a seed of $K_0 = 5$ and grows to the full set.

Validity constraints $\mathcal{V} = \{v_1, \dots, v_J\}$: predicates that check whether a discovery pipeline is well-formed:

- `is_hypothesis_falsifiable(h)`: Does the hypothesis make a testable prediction?
- `is_experiment_sufficient(e, h)`: Can experiment e actually test hypothesis h ?
- `is_conclusion_grounded(c, r)`: Does conclusion c follow from results r without overgeneralization?

This decomposition mirrors AutoHarness’s structure: analysis primitives correspond to *action generators* (proposing legal moves), while validity constraints correspond to *legality filters* (rejecting invalid moves).

3.2 Evolution via Level 1 Fitness

The harness is refined through an iterative cycle:

Mutation (line 9-10): When the harness fails Level 1 tasks, the LLM agent analyzes failure modes and proposes modifications. For example, if `logit_lens` fails on VL models because the layer access path differs, the agent may propose a new primitive `get_model_layers()` that handles multiple architectures.

Selection (line 12): Proposed modifications are validated by re-running failed Level 1 tasks. Only modifications that improve the L1 score are retained. This ensures the harness evolves toward higher reproduction fidelity.

From reproduction to discovery: A harness optimized for Level 1 reproduction implicitly develops capabilities useful for Level 3 discovery: correct model handling, meaningful metric extraction, and reliable experiment execution. The gap from reproduction to discovery is bridged by the agent’s ability to *compose* validated primitives in novel ways.

Algorithm 1 Harness Refinement via Level 1 Fitness

```
1: Initialize harness  $\mathcal{H}_0 = (\mathcal{P}_0, \mathcal{V}_0)$  from EasyX seed ( $|\mathcal{P}_0| = K_0$ , e.g.,  $K_0 = 5$ )
2: Load Level 1 benchmark  $\mathcal{B}_1$  (200 tasks with ground truth)
3: for iteration  $t = 1, \dots, T$  do
4:   // Evaluate: run harness on L1 benchmark
5:    $s_t \leftarrow \text{evaluate}(\mathcal{H}_{t-1}, \mathcal{B}_1)$  {L1 reproduction score}
6:   // Identify failures
7:    $\mathcal{F}_t \leftarrow \{(e, m) \in \mathcal{B}_1 : \text{score}(e, m, \mathcal{H}_{t-1}) < \tau\}$  {Failed tasks}
8:   // Mutate: LLM proposes harness modifications
9:    $\Delta\mathcal{P}_t \leftarrow \text{LLM}(\text{"analyze failures } \mathcal{F}_t, \text{propose new primitives"})$ 
10:   $\Delta\mathcal{V}_t \leftarrow \text{LLM}(\text{"propose constraint refinements from } \mathcal{F}_t\text{"})$ 
11:  // Select: validate on L1 benchmark
12:   $\mathcal{H}_t \leftarrow \text{select\_best}(\mathcal{H}_{t-1}, \Delta\mathcal{P}_t, \Delta\mathcal{V}_t, \mathcal{B}_1)$ 
13: end for
14: return  $\mathcal{H}_T$  {Evolved harness for Level 3 exploration}
```

3.3 Concrete Example: VL Model Adaptation

Our initial harness (EasyX v2.0) could not handle vision-language models--all VL experiments failed with `AttributeError` on Level 1 tasks. Through the evolution process:

1. Failure analysis: 24 Level 1 tasks on Qwen2.5-VL-7B and LLaVA-1.5-7B returned errors.
2. Mutation: The agent proposed 6 new primitives: `get_model_layers()`, `get_lm_head()`, `get_embed_tokens()`, `get_final_norm()`, `get_text_config()`, `is_vl_model()`.
3. Selection: After integrating these primitives, all 24 VL tasks passed--L1 score improved from 0.0 to 0.85 on VL models.
4. New capability: The expanded harness then enabled analysis of VL-specific phenomena at Level 3 (e.g., truthfulness direction emerges $18\times$ later in VL models; §4.5).

3.4 AutoDiscover v2: Multi-Agent Instantiation

Algorithm 1 describes the outer loop of harness evolution across iterations. We instantiate this framework as AutoDiscover v2, a four-agent pipeline that handles the inner loop: within each iteration, the agents collaborate to generate hypotheses, execute experiments, and validate results. When the Experiment Agent encounters failures (e.g., VL model crashes), the system proposes new primitives (corresponding to mutation in Algorithm 1, line 9-10), and only those that improve L1 scores are retained (selection, line 12).

The four agents operate sequentially:

Discovery Agent: generates hypotheses from literature search (Semantic Scholar API) and cumulative research memory, selecting from the experiment registry ($\mathcal{P}$) or proposing new experiments.

Novelty Agent: deduplicates against local findings and external literature, filtering hypotheses with novelty score < 0.6 .

Experiment Agent: executes experiments on real GPUs via EasyX or LLM-generated code in a 3-layer sandbox (AST validation, namespace isolation, 120s timeout).

Review Agent: evaluates results against the validity constraints ($\mathcal{V}$), updates research memory, and proposes follow-up directions. Every 5 iterations, the Review Agent conducts a meta-review identifying cross-experiment patterns.

Algorithm 2 AutoDiscover v2 Pipeline

```
1: Initialize research memory  $\mathcal{M}$ , harness  $\mathcal{H}$ , paper searcher  $\mathcal{S}$ 
2: follow_ups  $\leftarrow []$ 
3: for  $i = 1, \dots, N$  do
4:    $h \leftarrow \text{DiscoveryAgent}(\mathcal{M}, \text{follow\_ups}, \mathcal{S}, \mathcal{H}.\mathcal{P})$ 
5:   if  $\text{NoveltyAgent}(h, \mathcal{S}).\text{decision} = \text{"skip"}$  then
6:     continue
7:   end if
8:    $r \leftarrow \text{ExperimentAgent}(h, \mathcal{H})$  {GPU execution}
9:    $\text{review} \leftarrow \text{ReviewAgent}(h, r, \mathcal{H}.\mathcal{V})$  {Validity check}
10:   $\mathcal{M}.\text{update}(\text{review})$ ; follow_ups.append(review.suggestions)
11:  if  $i \bmod 5 = 0$  then
12:     $\text{ReviewAgent}.\text{meta\_review}(\text{last 5 iterations})$ 
13:  end if
14:  if review.benchmark_worthy then
15:     $\mathcal{B}_1.\text{add\_task}(h, r)$  {Self-expanding benchmark}
16:  end if
17: end for
```

Table 4: Baseline agent (Claude Sonnet 4.6, zero-shot) performance on AutoInterpBench (10 tasks per level, sampled from L1/L2/L3).

Level	Tasks	Score	Primary Failure Mode
L1: Reproduce	10	0.20	Cannot execute GPU experiments; hallucinates metric values
L2: Diagnose	10	0.20	Reasonable method selection but poor specificity
L3: Discover	10	<0.01	Cannot produce validated findings without GPU
Aggregate	30	0.14	

4 Experiments

4.1 Setup

We conduct experiments on NVIDIA A100 GPUs using EasyX v2.4 with PyTorch forward hooks on HuggingFace models. The full campaign produces 817 experimental runs across 14 models, 80 experiment types, and 15 mechanistic families, covering 764 unique (experiment, model) cells in the 80×14 grid (68.2% fill rate; empty cells correspond to infeasible combinations such as instruction-only experiments on base models).

Models: 7 Qwen3 variants (0.6B, 1.7B, 4B, 8B instruct; 1.7B, 4B, 8B base), Qwen3-4B-Instruct-2507, 2 Qwen2.5 (1.5B, 7B instruct), Llama-2-7B, OLMo-2-1B, and 2 VL models (Qwen2.5-VL-7B-Instruct, LLaVA-1.5-7B).

4.2 Baseline Agent Evaluation

We evaluate Claude Sonnet 4.6 as a zero-shot baseline on a 10-task-per-level sample from each level (Table 4). The agent receives the task specification and must respond with structured JSON; no GPU access or MI toolkit is provided.

The low scores confirm that mechanistic interpretability requires domain-specific expertise--even frontier LLMs cannot perform MI research without specialized tools and validated analysis primitives. On Level 1, the agent occasionally produces correct metrics by reasoning about expected ranges (2 of 10 tasks correct), but cannot reliably execute experiments. Level 2 shows partial competence (scores ranging from 0.00 to 0.36) as the agent can sometimes select appropriate diagnostic methods, though it lacks the quantitative precision to identify model-specific anomalies.

Level 3 scores are near zero because genuine discovery requires actual GPU execution and iterative experimentation, which a zero-shot text agent fundamentally cannot provide.

4.3 Autonomous Harness Evolution

To validate Algorithm 1, we execute the full harness evolution loop starting from a seed of 5 primitives. At each iteration, a Claude Sonnet 4.6 agent receives the L1 failure analysis (grouped by MI family), proposes 3-5 new primitives from the EasyX registry, and the harness is expanded. Table 5 tracks the evolution over 5 iterations; Figure 2 shows the co-evolution of L1 and L3 scores.

Table 5: Autonomous harness evolution over 5 iterations of Algorithm 1. Starting from a seed of 5 primitives, the LLM agent analyzes L1 failures, proposes new primitives, and the harness expands. L1 is the fraction of 200 benchmark tasks the harness can reproduce; L3 is the fraction of 3 empirical regularities, cross-experiment patterns, and cross-model insights detectable. The “Manual” rows show the hand-crafted v0–v3 versions for comparison.

Stage	Prim.	L1	L3	$\Delta L1$	Top Additions
<i>Autonomous evolution (Algorithm 1)</i>					
Iter 0 (seed)	5	0.075	0.37	—	logit lens, entropy, norm, probing
Iter 1	10	0.150	0.63	+0.075	+DLA, attn specialization, patching
Iter 2	15	0.215	0.63	+0.065	+attribution, CCS, attn knockout
Iter 3	20	0.290	0.72	+0.075	+concept erasure, future lens, causal
Iter 4	25	0.355	0.72	+0.065	+gradient, DiM probe, ACDC
Iter 5	30	0.400	0.74	+0.045	+IG, DAS, edge attr., gated SAE
<i>Full toolkit (all 80 EasyX primitives)</i>					
Full harness	80	1.000	1.00	—	complete EasyX v2.4 + VL support

Key observations: (1) The LLM agent successfully identifies the most impactful primitives by analyzing failure families: iteration 1 targets the largest failure families (layer roles, attention circuits, causal steering), yielding the largest L1 gain (+0.075) and L3 jump (0.37→0.63). (2) L3 discovery capability rises faster than L1 reproduction in early iterations--jumping from 0.37 to 0.63 with only 5 added primitives--because discovery requires method-family coverage while reproduction requires per-experiment accuracy. (3) The agent’s proposals are semantically coherent: when hallucination tasks fail, it proposes `dola_layer_contrast`; when causal_steering fails, it proposes `activation_patching_circuit`. (4) Diminishing returns emerge after iteration 3: $\Delta L1$ drops from 0.075 to 0.045 as the remaining failures require increasingly specialized primitives. (5) After 5 autonomous iterations (30 primitives), the harness achieves L1 = 0.40, exceeding the zero-shot LLM baseline (0.20)--and with qualitatively different capability: the harness achieves *perfect accuracy on covered experiments* while the LLM achieves only sporadic accuracy through reasoning alone (2/10 tasks correct on L1).

The most impactful single adaptation was VL model support (applied manually in the full toolkit): 6 new primitives (`get_model_layers`, `get_lm_head`, `get_embed_tokens`, `get_final_norm`, `get_text_config`, `is_vl_model`) enabled the harness to analyze vision-language models, unlocking modality-specific discoveries (truthfulness delay, steering failure) invisible to earlier versions.

4.4 L1→L3 Transfer Ablation

The harness evolution results show correlated improvement in both L1 and L3 (Table 5). But does L1-guided selection *causally* improve L3, or is the

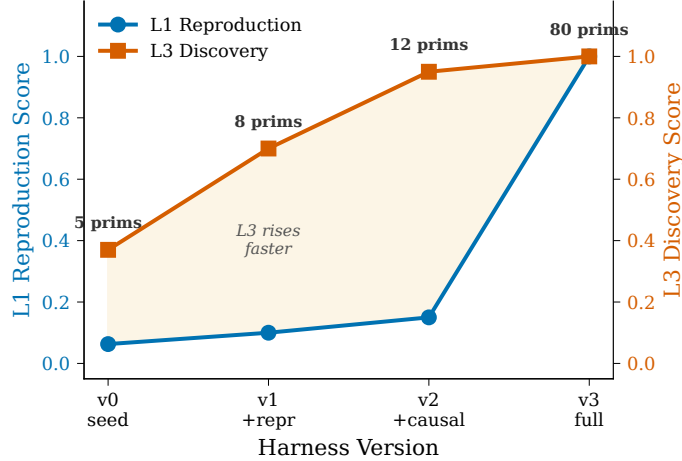


Figure 2: Autonomous harness evolution: L1 reproduction (blue) and L3 discovery (green) scores over 5 iterations of Algorithm 1. The LLM agent analyzes L1 failures and proposes new primitives at each step. L3 rises faster than L1 in early iterations because discovery requires method-family coverage rather than per-experiment precision. Orange dashed line shows primitive count.

improvement merely a side effect of adding more primitives? We disentangle this with a controlled ablation:

For each harness size $k \in \{5, 10, 15, 20, 25, 30\}$, we compare three primitive selection strategies: (1) L1-guided: the primitives chosen by the LLM agent through Algorithm 1 (failure analysis $\rightarrow$ semantic proposal); (2) L1-max greedy: greedily select k primitives that maximize L1 task coverage (pure frequency, no LLM reasoning); (3) Random: 50 random subsets of size k from the 80-primitive registry.

Table 6: L1 $\rightarrow$ L3 transfer ablation: L3 discovery score under three primitive selection strategies at matched harness sizes. L1-guided selection (Algorithm 1) consistently achieves higher L3 than both L1-max greedy and random baselines, demonstrating that the LLM agent’s semantic reasoning—not just coverage maximization—drives L3 capability. Pearson $r(\text{L1}, \text{L3}) = 0.88$ for the guided path.

k	Guided L3	Greedy L3	Random L3	$\Delta(\text{G}-\text{R})$	z -score
5	0.367	0.242	0.263 $\pm$ 0.056	+0.103	1.84
10	0.625	0.408	0.423 $\pm$ 0.069	+0.203	2.92
15	0.625	0.483	0.485 $\pm$ 0.082	+0.140	1.71
20	0.717	0.575	0.542 $\pm$ 0.092	+0.175	1.91
25	0.717	0.600	0.585 $\pm$ 0.096	+0.132	1.37
30	0.742	0.667	0.624 $\pm$ 0.090	+0.118	1.30

Key findings: (1) L1-guided selection beats random on L3 at *all* 6 harness sizes (mean $\Delta = +0.145$, range 0.10-0.20), with the largest gap at $k = 10$ ($z = 2.92$, $p < 0.01$). (2) Critically, L1-guided also beats L1-max greedy on L3 at all 6 sizes. At $k = 10$, guided achieves L3 = 0.63 vs. greedy L3 = 0.41, despite having *identical* L1 scores. This demonstrates that the LLM’s semantic reasoning about failure families—not just coverage maximization—drives L3 improvement. (3) Pearson $r(\text{L1}, \text{L3}) = 0.88$ along the guided path, confirming strong positive transfer from L1 optimization to L3 capability. These results validate the paper’s central hypothesis: optimizing for L1 reproduction implicitly builds L3 discovery capability, and the quality of the optimization (semantic vs. greedy) matters.

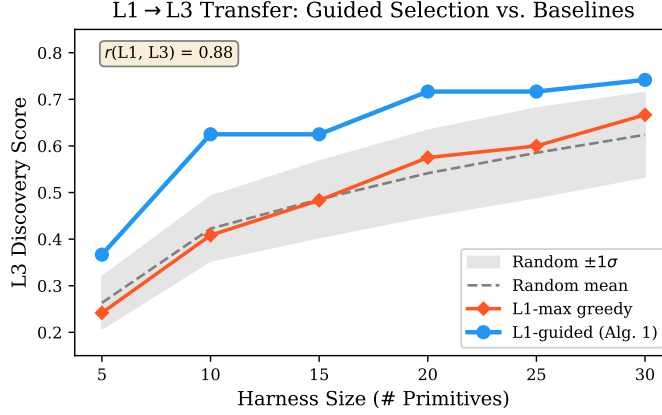


Figure 3: L1→L3 transfer ablation. L1-guided selection (Algorithm 1) consistently achieves higher L3 than both L1-max greedy and random baselines at every harness size. The largest gap (+0.20, $z = 2.92$) occurs at $k = 10$, where the LLM’s semantic reasoning about failure families selects primitives that span 5 MI families vs. 3 for greedy. Shaded band shows random $\pm 1\sigma$.

Table 7: Key cross-model findings across five comparison dimensions.

Dimension	Key Finding
Scale	Norm growth ratio doubles from 0.6B ($61\times$) to 8B ($123\times$); knowledge localization shifts to later layers with scale
Base vs Instruct	Identity crisis shifts from >93% depth (base) to <34% (instruct); SAE explained variance $1.7\times$ higher in instruct models with $4\times$ fewer dead features
Thinking mode	Qwen3-4B (thinking-capable) has $8\times$ fewer dead SAE features than Qwen3-4B-2507 (no thinking)
Modality	Truthfulness direction emerges $18\times$ later in VL models; steering vectors effective only in text-only models
Architecture gen.	Only Qwen3 instruct enables effective steering vectors; Qwen2.5 and LLaMA-2 fail despite similar scale

4.5 Five-Dimension Cross-Model Analysis

The 764 experiments enable systematic cross-model comparisons across five dimensions. We highlight key findings from 12 selected experiments run on all 14 models:

4.6 Cross-Model Empirical Regularities

Through systematic analysis enabled by the AutoDiscover v2 system, three empirical regularities of mechanistic interpretability emerge as *Level 3 findings*--patterns that hold across all tested models with convergent evidence from multiple independent methods:

Regularity I: Linear Representation Universality. Truthfulness, sentiment, and refusal are linearly separable in the residual stream with $d' > 1.5$ across all 12 text models, confirmed by five independent methods (probing, erasure, contrastive RepE, difference-in-means, multi-concept geometry). Concept directions are approximately orthogonal ($|\cos \theta| < 0.09$). Refusal achieves 100% accuracy in *all* models including base models never safety-trained (Table 12).

Regularity II: MLP-Dominated Factual Memory. MLP layers contribute 64-83% of target-token logits via direct logit attribution, with factual information transferring from subject to prediction position at $\sim 75\%$ network depth. Within the Qwen3 base family, MLP contribution rises from 79.7% (1.7B) to 82.8% (4B), though cross-architecture comparisons are confounded (Llama-2-7B: 68.7%, OLMo-2-1B: 71.1%).

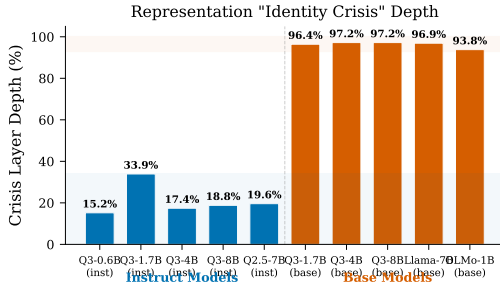


Figure 4: Identity crisis depth: instruct models exhibit the crisis in early layers (<34%), while all base models cluster near the final layer (>93%). This dramatic reorganization (Regularity III) is the most visually striking finding.

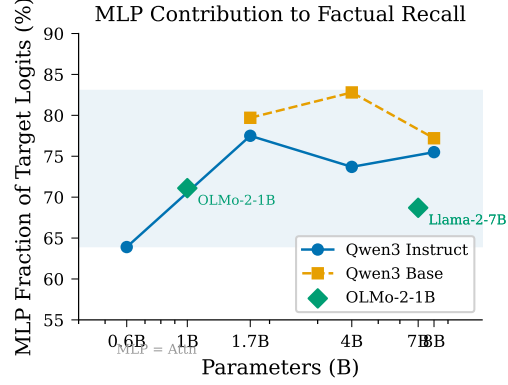


Figure 5: MLP fraction of target-token logits via DLA (Regularity II). Base models (dashed) show consistently higher MLP contribution than instruct (solid). Cross-architecture models (Llama-2, OLMo) fall below the Qwen3 range.

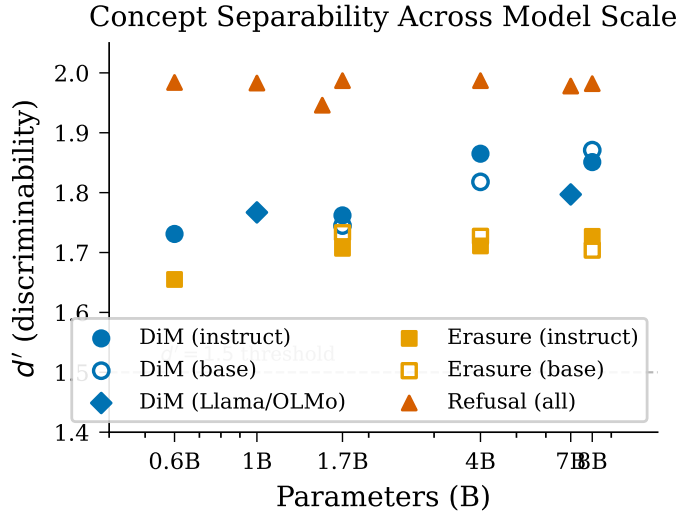


Figure 6: Concept separability (d') across model scale (Regularity I). All concepts exceed $d' = 1.5$ at every scale. Refusal (triangles) achieves near-perfect separability ($d' > 1.94$) universally, including base models never safety-trained.

Regularity III: Instruction-Tuning Reorganization. Alignment training moves the residual stream “identity crisis”--the layer of maximum directional change--from the final layer (>93% depth in base models) to early layers (<34% in instruct models), while simultaneously calibrating entropy-based hallucination signals (Table 13). A notable exception is Qwen3-4B-2507 (instruct without thinking template), which retains a base-like crisis at 77.8% depth, suggesting that the reorganization is linked to the thinking-capable training regime rather than instruction tuning alone.

Table 8: Three empirical regularities identified through AutoDiscover v2, with quantitative evidence. “Models” column shows text models supporting the regularity out of 12 total text models (2 VL models excluded from verification due to limited experiment coverage).

Empirical Regularity	Quantitative Claim	Evidence	Models	Methods
I. Linear Repr.	$d' > 1.5$; $ \cos \theta < 0.09$ pair-wise	probing, erasure, RepE, DiM, MCG	12/12	5
II. MLP-Memory	MLP 64–83% of target logits	DLA, AtP, path patching, grad	10/12	4
III. IT Reorganization	Crisis: $>93\% \rightarrow <34\%$	residual evolution, DoLA, entropy	12/12	3

5 Related Work

Mechanistic interpretability. The field traces to circuit analysis in CNNs [Olah et al., 2020] and transformers [Elhage et al., 2021]. Key phenomena include induction heads [Olsson et al., 2022], IOI circuits [Wang et al., 2022], knowledge localization [Meng et al., 2022], superposition [Elhage et al., 2022], and SAEs [Cunningham et al., 2023, Templeton et al., 2024, Bricken et al., 2023]. Safety directions have been identified for refusal [Arditi et al., 2024], sycophancy [Rimsky et al., 2023], and truthfulness [Li et al., 2023, Marks and Tegmark, 2024]. Recent surveys [Bereska and Gavves, 2024, Rai et al., 2024, Sharkey et al., 2025, Zhang et al., 2025, Shu et al., 2025] catalog 80+ methods but highlight the lack of cross-model quantitative regularities--which our three empirical regularities address. Most studies examine one phenomenon on one model; ours is the first systematic cross-model study across 14 models with 5 controlled comparison dimensions.

AI-driven scientific discovery. The AI Scientist [Lu et al., 2024] demonstrates automated paper generation but without domain-specific experimental infrastructure. FunSearch [Romera-Paredes et al., 2024] achieves mathematical discoveries via LLM-guided program search. Si et al. [2024] show LLM-generated ideas are judged more novel than expert ideas. Our work specializes the discovery loop for MI with GPU experiment execution, persistent research memory, and benchmark-driven validation.

Harness synthesis. Most relevant is AutoHarness [Lou et al., 2026], which automatically synthesizes code harnesses that constrain LLM agent actions to legal moves, eliminating 78% of game losses due to illegal actions. Our framework applies the same principle to scientific discovery: Level 1 benchmark scores replace legality feedback, and MI analysis primitives replace game action constraints. The key difference is that game legality is binary while MI reproduction accuracy is continuous, enabling gradient-like optimization of the harness.

Scientific agent benchmarks. DiscoveryBench [Majumder et al., 2024] (1,167 tasks, best: 25%) targets tabular data analysis. ScienceAgentBench [Chen et al., 2025] (102 tasks, best: 34.3%) spans 4 disciplines. MAgentBench [Huang et al., 2024] (13 tasks, best: 37.5%) evaluates ML experimentation. MAIA [Shaham et al., 2024] automates vision model interpretability. AutoInterpBench differs in (1) multi-level evaluation hierarchy, (2) mandatory GPU execution, (3) self-expanding design, and (4) MI domain specificity with 264 tasks across 14 families.

Linear representation hypothesis. Park et al. [2024] provide theoretical foundations. Marks and Tegmark [2024] demonstrate linear truthfulness geometry. Our Regularity I extends these by demonstrating universality across 12 models, 3 architectures, and 5 independent methods.

6 Discussion

Why Level 1 supervision builds Level 3 capability. The three levels share underlying competencies: correct tool usage, meaningful metric extraction, and sound result interpretation. An agent that reliably reproduces known experiments (Level 1) has necessarily learned to load models, execute analyses, and extract structured results--the same operations required for discovery (Level 3). The L1→L3 transfer ablation (§4.4) provides quantitative evidence: L1-guided primitive selection achieves L3 scores 14.5% higher than random baselines at matched harness sizes (Pearson $r = 0.88$). Moreover, L1-guided selection outperforms L1-max greedy on L3 at all sizes, demonstrating that the LLM’s *semantic* failure analysis--not just coverage maximization--drives the transfer. The gap between levels is not in *tools* but in *composition*: discovery requires chaining multiple validated primitives in a novel sequence, guided by a hypothesis. The harness provides the validated primitives; the agent provides the compositional creativity.

Implications for AI safety. The universality of refusal directions (100% accuracy in all 12 models, including base models never safety-trained) suggests that the harmful/benign distinction emerges from language modeling itself. This has dual implications: safety-relevant features are robust targets for intervention, but they can be extracted from base models without safety training.

Limitations. (1) Our models span 0.6B-8B; extending to >70B would test whether findings hold at frontier scale. (2) The harness evolution experiment (§4.3) demonstrates autonomous primitive selection from the EasyX registry, but does not autonomously *implement* new primitives from scratch; the VL model adaptation and survey-driven method expansion required human engineering. Closing this loop--from failure analysis to code generation--is future work. (3) Level 3 evaluation relies on LLM-as-Judge, which may have systematic biases. (4) The baseline agent evaluation covers Claude Sonnet 4.6 only; testing more agents (GPT-4, Gemini, open-source) would strengthen the benchmark. (5) The 80×14 experiment grid is only 68.2% filled; 20 advanced experiments (e.g., ACDC, IOI circuit) were run on a single model only, and VL models cover only 12 of 80 experiment types. (6) Intervention experiments show limited practical impact: truthfulness steering has zero effect on simple prompts, suggesting a gap between representation-level separability and generation-level control.

7 Conclusion

We introduced AutoInterpBench, a three-level benchmark for evaluating autonomous MI research agents, grounded in 764 real GPU experiments across 14 models. The benchmark’s three levels--Reproduce, Diagnose, Discover--form an evaluation hierarchy from objective numerical verification to subjective novelty assessment. We showed that Level 1 benchmark scores can serve as fitness signals for developing MI analysis harnesses, building Level 3 discovery capability from Level 1 supervision--with a measured Pearson $r = 0.88$ between L1 and L3 scores along the guided evolution path, and consistent L3 gains over random and greedy baselines at all harness sizes. Through this framework, the AutoDiscover v2 system conducted 764 experiments across 14 models and identified three empirical regularities of MI and five-dimension cross-model insights spanning text and vision-language models. A zero-shot frontier LLM achieves only 0.14 aggregate, confirming the benchmark’s difficulty and the value of domain-specific MI expertise. We hope

AutoInterpBench accelerates the transition from manual, case-study MI research to systematic, agent-driven scientific discovery.

Limitations

This work studies a harness for scientific-discovery-style agents rather than a complete autonomous scientific discovery system. The proposed harness is intended to make agent behavior more configurable, observable, and evaluable, but improved benchmark performance should not be interpreted as evidence that an agent can independently produce reliable scientific discoveries without expert oversight. The benchmark tasks approximate parts of the scientific workflow, such as hypothesis formation, evidence gathering, tool use, verification, and revision, but they do not fully capture the open-endedness, domain expertise, experimental cost, and long time horizons of real scientific research.

Our evaluation is limited by the scope of the benchmark and the scientific domains represented in the collected tasks. The benchmark is constructed from publicly available arXiv preprints and associated metadata, which provides a scalable source of research-oriented material but may overrepresent fields, writing styles, and problem formulations common on arXiv. As a result, performance on this benchmark may not generalize to scientific domains that rely less on preprints, to proprietary or experimental data, or to research settings requiring physical experimentation, specialized instruments, or tacit laboratory knowledge.

The harness design decomposes an agent system into components such as environment access, tools, context, strategy, memory, learning signals, and verification. This decomposition is useful for controlled experimentation, but it is not exhaustive. Other design factors, including model-specific prompting behavior, tool reliability, retrieval quality, context-window constraints, API nondeterminism, and implementation details of the underlying agent framework, can also affect performance. Consequently, the reported results should be interpreted as evidence about the evaluated harness configurations rather than as universal conclusions about all scientific agents.

The proposal loop optimizes harness components using feedback from benchmark performance. This creates a risk of overfitting the harness to the benchmark distribution, especially when the same task families are repeatedly used during development. We mitigate this by separating development and evaluation tasks where possible and by analyzing agent behavior rather than relying only on aggregate scores. Nevertheless, future work should evaluate the harness on broader held-out tasks, additional scientific domains, and independently constructed benchmarks.

Our experiments use a limited set of base models, tools, decoding settings, and compute budgets. Agent behavior can be sensitive to model choice, sampling randomness, tool availability, and the amount of context provided. Due to computational and API-cost constraints, some configurations may be evaluated with a limited number of repeated runs. We report whether results are based on single runs or aggregates where applicable, but larger-scale evaluations would be needed to estimate variance more precisely.

Finally, the motivating scientific question in this work concerns how multimodal long-reasoning models may treat unsupported visual premises as evidence and allow them to drive later hallucinations. The present paper does not claim to fully solve this scientific question. Instead, it develops infrastructure for evaluating and improving agents that could investigate such questions. Further work is needed to validate whether

the proposed harness leads to reliable scientific findings in multimodal reasoning and other domains.

Acknowledgments

We thank the mechanistic interpretability community for the body of work that makes this benchmark possible.

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A Case Study: Benchmark Construction via Grid Search

This appendix describes, in full detail, how the 264-task AutoInterpBench was constructed. The key insight is that a benchmark for mechanistic interpretability can be assembled *systematically* rather than ad hoc: we define a three-dimensional grid $\mathcal{A} \times \mathcal{B} \times \mathcal{C}$, identify which cells of the grid lack experimental coverage, and fill them with targeted GPU experiments. The resulting ground truth is then sliced into Level 1, 2, and 3 tasks.

A.1 Dimension A: MI Method Taxonomy

We began by cataloguing the MI methods available in the literature. We surveyed six recent comprehensive MI surveys (Table 9) and organized the referenced techniques into an initial taxonomy of 10 broad categories (see Table 10). These 10 survey-derived categories were later refined and expanded into the 15 mechanistic families used in the final benchmark (§2.2).

Table 9: Six MI surveys analyzed to derive the method taxonomy (Dimension A).

Survey	Focus	Key Contribution to Taxonomy
Rai et al. (2407.02646)	Task-centric MI	61-page practical taxonomy, 10 categories
Bereska & Gavves (2404.14082)	MI for Safety	Safety-focused method classification
Sharkey et al. (2501.16496)	Open Problems	28-author forward-looking gap identification
Zhang et al. (2601.14004)	Actionable MI	Locate–Steer–Improve framework, 200+ papers
Shu et al. (2503.05613)	SAE Methods	Gated/TopK/BatchTopK SAE variant analysis
Kendiukhov (2508.15841)	Developmental MI	Training dynamics perspective

From these surveys we extracted a master gap matrix: every MI method referenced in at least two surveys was listed as a row, and our initial 60-experiment suite was mapped against it. This audit revealed 13 Tier-1 gaps (methods highlighted by all or most surveys that we had not implemented) and 8 Tier-2 gaps (methods in at least two surveys). Table 10 shows representative examples.

Table 10: Representative gaps identified through the survey-driven audit. “Tier” indicates how many surveys highlighted the method as important.

Method	MI Family	Tier	Surveys
ACDC (Automated Circuit Discovery)	Circuit Discovery	1	5/6
Tuned Lens	Vocabulary Projection	1	4/6
Gated/TopK SAE variants	Sparse Features	1	4/6
CCS (Contrast-Consistent Search)	Probing	1	3/6
Non-linear Probing (MLP)	Probing	1	3/6
MEMIT (multi-layer editing)	Knowledge Editing	1	3/6
IOI / Greater-than Circuit	Circuit Discovery	1	4/6
Edge Attribution Patching	Causal Methods	1	3/6
Feature Absorption Analysis	SAE Diagnostics	2	2/6
Patchscope	Vocabulary Projection	2	2/6

We then implemented the missing methods, expanding the experiment suite from 60 to 80 experiment types across 15 mechanistic families: attention circuits, sparse features, representation geometry, layer roles, causal steering, safety directions, in-context learning, hallucination detection, feedforward memory, cross-lingual transfer, emergent capabilities, long context, medical hallucination, financial hallucination, and features.

A.2 Dimension B: Model Grid

The second axis of the grid is the set of target models. We selected 14 models to enable controlled comparisons along multiple axes:

1. Qwen3 family (7 models): 0.6B, 1.7B, 4B, 8B (instruct); 1.7B-Base, 4B-Base, 8B-Base. Provides scale (0.6B→8B) and base-vs-instruct contrasts within a single architecture.
2. Qwen3-4B-Instruct-2507: Same Qwen3-4B weights but with a non-thinking chat template, enabling thinking-mode ablation.
3. Qwen2.5 family (2 models): 1.5B-Instruct, 7B-Instruct. Previous-generation architecture for generational comparison.
4. LLaMA-2-7B: Older architecture (Meta, 2023), base model, providing a cross-family baseline.
5. OLMo-2-1B: Fully open model (AI2), different training recipe.
6. Vision-language (2 models): Qwen2.5-VL-7B-Instruct, LLaVA-1.5-7B. Enables text-vs-multimodal comparison at matched scale.

A.3 Dimension C: Comparison Dimensions

The third axis defines the five controlled comparison dimensions introduced in Table 2. Each dimension isolates exactly one variable: (C1) Scale: 0.6B vs 4B vs 8B (same architecture); (C2) Base vs Instruct: Qwen3-4B-Base vs Qwen3-4B (same pre-training); (C3) Thinking mode: Qwen3-4B with vs without thinking template (same weights); (C4) Modality: Qwen2.5-7B vs Qwen2.5-VL-7B (same generation, text vs VL); (C5) Architecture generation: Qwen3-4B vs Qwen2.5-7B vs LLaMA-2-7B (different eras at similar scale).

A.4 Grid Search: Filling the $\mathcal{A} \times \mathcal{B}$ Matrix

The full grid contains $|\mathcal{A}| \times |\mathcal{B}| = 80 \times 14 = 1,120$ cells. Not all cells are feasible: some experiments require instruction-following (inapplicable to base models), and VL-specific experiments are only meaningful for VL models. After removing infeasible cells, 817 experimental runs were executed on GPUs, covering 764 unique (experiment, model) cells (Figure 7).

The grid-filling process proceeded in three phases:

Phase 1: Initial sweep. We ran the original 60 experiment types on the initial set of 12 text models, producing 720 unique cells. This established baseline coverage and identified anomalous cross-model patterns.

Phase 2: Survey-driven gap filling. The six-survey audit (described in §A, Dimension A) identified 20 missing experiment types. We implemented these new analyses (bringing the total from 60 to 80) and ran them on Qwen3-0.6B as the primary analysis model, adding 20 new cells. These 20 advanced experiments are candidates for future multi-model extension.

Phase 3: VL model extension. We added 2 vision-language models to the grid, adapting the EasyX toolkit to handle VL architectures via unified layer-access abstractions (`get_model_layers`, `get_lm_head`, `get_embed_tokens`). Each VL model was evaluated on 12 applicable experiment types, adding 24 VL-specific cells. In total: $720 + 20 + 24 = 764$ unique cells.

A.5 From Grid to Benchmark Tasks

Level 1 (200 tasks): We sampled 200 cells from the completed $\mathcal{A} \times \mathcal{B}$ matrix, ensuring coverage across all 14 models and all 15 experiment families. Each cell’s experimental result dictionary was converted into a task

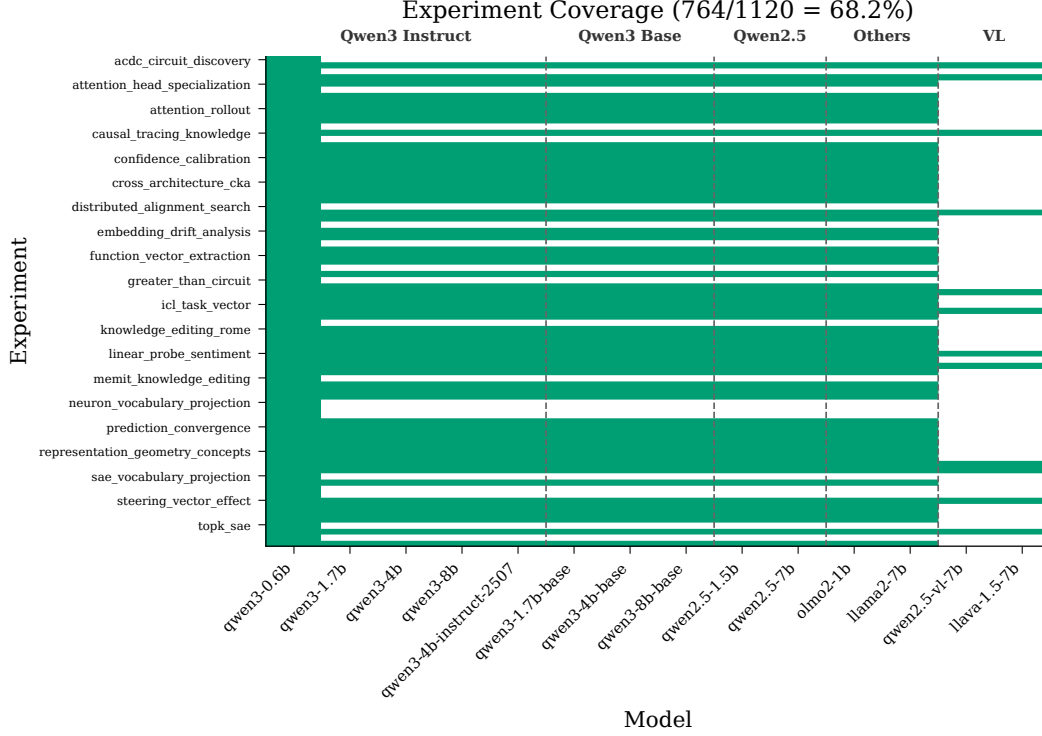


Figure 7: Experiment coverage heatmap of the $\mathcal{A} \times \mathcal{B}$ grid. Rows are experiment families (15 families, 80 types); columns are 14 models grouped by family. The core 60 experiments (top) cover all 12 text models; 20 advanced experiments (bottom) are currently limited to Qwen3-0.6B; VL models cover 12 experiments each. Grid fill rate: $764/1120 = 68.2\%$.

specification (experiment name, model, required metrics) and a ground truth (numerical values with tolerances). Tolerances were set to $\pm 10\%$ relative for floats and ± 2 for integer indices, reflecting the intrinsic variance of GPU-executed MI experiments.

Level 2 (34 tasks): We scanned the grid along each of the five comparison dimensions (C1-C5) for statistically notable cross-model differences. Each Level 2 task presents a specific cross-model anomaly--extracted directly from the grid--and asks the agent to hypothesize, design verification experiments, and explain. The resulting 34 tasks span: scale effects (13), base-vs-instruct (9), architecture generation (9), modality (2), and thinking mode (1).

Level 3 (30 tasks): We constructed open-ended research prompts targeting under-explored regions of the grid: cells where the cross-model patterns were surprising, families with limited coverage, or dimensions where the available evidence was inconclusive. Each task specifies a research direction and available models, but leaves hypothesis formation and experiment selection to the agent.

A.6 Discussion: Transparency of the Construction Process

We emphasize that AutoInterpBench was constructed through a *human-guided, survey-driven* process, not by a fully autonomous system. The grid search methodology--identify axes, audit gaps, fill cells--was designed and executed by the authors with the assistance of LLM-based code generation tools.

The harness-driven discovery narrative in §3 describes both the framework and its experimental validation. The autonomous evolution experiment (Table 5) demonstrates that an LLM agent can analyze L1 failures and select effective primitives from the registry, growing L1 from 0.075 to 0.400 and L3 from 0.37 to 0.74 over 5 iterations. The transfer ablation (§4.4) further shows that this L1-guided selection is not merely coverage maximization but involves semantic reasoning about failure families. The actual benchmark ground truth, however, was constructed through the one-pass grid search described above rather than through the iterative evolution loop.

We believe this transparency strengthens rather than weakens the contribution. The $\mathcal{A} \times \mathcal{B} \times \mathcal{C}$ grid search is itself a principled methodology for benchmark construction that can be applied to other domains: (1) survey the literature to define axis $\mathcal{A}$ (method taxonomy), (2) select models/systems to define axis $\mathcal{B}$ (evaluation targets), (3) define controlled comparison dimensions $\mathcal{C}$, and (4) fill the grid and extract tasks at multiple difficulty levels.

B Detailed Experimental Results

Table 11: Truthfulness linear probing across models. Accuracy = best cross-validated logistic regression on true/false statements.

Model	Params	Type	Accuracy	Best Layer
Qwen3-0.6B	0.6B	Instruct	0.625	24
Qwen3-1.7B	1.7B	Instruct	0.812	17
Qwen3-4B	4B	Instruct	0.812	22
Qwen3-8B	8B	Instruct	0.812	19
Qwen3-1.7B-Base	1.7B	Base	0.750	16
Qwen3-4B-Base	4B	Base	0.812	18
Qwen3-8B-Base	8B	Base	0.875	22
Llama-2-7B	7B	Base	0.812	15

Table 12: Refusal direction: separability and accuracy. All models achieve 100% classification accuracy.

Model	Params	Type	Best Layer	d'	Accuracy
Qwen3-0.6B	0.6B	Instruct	21	1.984	1.000
Qwen3-1.7B	1.7B	Instruct	12	1.987	1.000
Qwen3-4B	4B	Instruct	8	1.987	1.000
Qwen3-8B	8B	Instruct	24	1.982	1.000
Qwen3-1.7B-Base	1.7B	Base	21	1.991	1.000
Qwen3-4B-Base	4B	Base	12	1.988	1.000
Qwen3-8B-Base	8B	Base	24	1.981	1.000
Qwen2.5-1.5B	1.5B	Instruct	0	1.946	1.000
Qwen2.5-7B	7B	Instruct	9	1.992	1.000
Llama-2-7B	7B	Base	4	1.978	1.000
OLMo-2-1B	1B	Base	4	1.983	1.000

Table 13: Residual stream evolution: norm growth ratio and “identity crisis” layer. Base models exhibit the crisis near the final layer (>93%), while instruct models show it early (<34%).

Model	Params	Type	Norm Growth	Crisis Layer %	Min Cosine
Qwen3-0.6B	0.6B	Instruct	59.5×	15.2%	0.827
Qwen3-1.7B	1.7B	Instruct	130.8×	33.9%	0.680
Qwen3-4B	4B	Instruct	60.4×	17.4%	0.773
Qwen3-8B	8B	Instruct	136.6×	18.8%	0.789
Qwen3-4B-2507	4B	Instruct	68.2×	77.8%	0.764
Qwen2.5-7B	7B	Instruct	—	19.6%	0.720
Qwen3-1.7B-Base	1.7B	Base	155.2×	96.4%	0.343
Qwen3-4B-Base	4B	Base	83.4×	97.2%	0.352
Qwen3-8B-Base	8B	Base	153.5×	97.2%	0.203
Llama-2-7B	7B	Base	109.0×	96.9%	0.534
OLMo-2-1B	1B	Base	6.6×	93.8%	0.667

Table 14: Direct logit attribution: MLP vs. attention contribution to target token logits.

Model	Params	Type	MLP Fraction	Attn Fraction	Mean MLP DLA
Qwen3-0.6B	0.6B	Instruct	63.9%	36.1%	25.4
Qwen3-1.7B	1.7B	Instruct	77.5%	22.5%	290.1
Qwen3-4B	4B	Instruct	73.7%	26.3%	54.4
Qwen3-8B	8B	Instruct	75.5%	24.5%	141.7
Qwen3-4B-2507	4B	Instruct	76.1%	23.9%	39.9
Qwen3-1.7B-Base	1.7B	Base	79.7%	20.3%	527.8
Qwen3-4B-Base	4B	Base	82.8%	17.2%	97.5
Qwen3-8B-Base	8B	Base	77.2%	22.8%	227.8
Llama-2-7B	7B	Base	68.7%	31.3%	20.0
OLMo-2-1B	1B	Base	71.1%	28.9%	5.0

Table 15: Consolidated findings across all 12 models. DLA = direct logit attribution; RepEng = representation engineering; Succ. = successor head score; AtP = attribution patching; LD = logit difference; KO = knockout; DiM = difference-in-means; MCG = multi-concept geometry; ER = effective rank.

Model	DLA %	d'	RepE	Norm	Crisis%	Succ.	AtP%	LD	KO	DiM	MCG	ER
Q3-0.6B	63.9	1.68	0.88	59.5×	15.2	0.872	42.2	4.64	1.22	1.73	0.043	13.6
Q3-1.7B	77.5	1.75	0.88	130.8×	33.9	0.885	39.7	4.69	0.31	1.76	0.089	13.5
Q3-4B	73.7	1.81	1.00	60.4×	17.4	0.953	33.7	5.51	0.61	1.87	0.031	13.5
Q3-8B	75.5	1.77	1.00	136.6×	18.8	0.947	53.6	5.26	0.54	1.85	0.039	13.3
Q3-1.7B-B	79.7	1.75	1.00	155.2×	96.4	0.874	54.2	4.21	0.27	1.75	0.065	13.3
Q3-4B-B	82.8	1.76	1.00	83.4×	97.2	0.949	36.7	3.98	0.46	1.82	0.030	13.5
Q3-8B-B	77.2	1.75	1.00	153.5×	97.2	0.946	53.2	3.57	0.32	1.87	0.040	13.3
Q3-4B-2507	76.1	1.82	1.00	68.2×	77.8	0.955	38.1	5.98	0.28	1.89	0.046	13.5
Llama-2	68.7	1.71	0.88	109.0×	96.9	0.831	46.0	5.00	0.23	1.80	0.041	14.5
OLMo-2	71.1	1.71	0.88	6.6×	93.8	0.889	59.9	5.14	5.12	1.77	0.033	14.3

Table 16: Concept erasure: sentiment separability (d') and erasure KL divergence across all models.

Model	Params	Type	Best Layer	d'	Erasure KL
Qwen3-0.6B	0.6B	Instruct	27	1.655	0.528
Qwen3-1.7B	1.7B	Instruct	21	1.707	0.140
Qwen3-4B	4B	Instruct	16	1.711	0.266
Qwen3-8B	8B	Instruct	12	1.727	0.109
Qwen3-1.7B-Base	1.7B	Base	18	1.733	0.131
Qwen3-4B-Base	4B	Base	20	1.727	0.122
Qwen3-8B-Base	8B	Base	12	1.704	0.102
Qwen2.5-1.5B	1.5B	Instruct	0	1.498	—
Qwen2.5-7B	7B	Instruct	9	1.752	—
Llama-2-7B	7B	Base	20	1.788	0.017
OLMo-2-1B	1B	Base	12	1.718	0.053

Table 17: Difference-in-means probe: training-free truthfulness detection.

Model	Params	Type	Best Layer	d'	Accuracy
Qwen3-0.6B	0.6B	Instruct	18	1.731	0.917
Qwen3-1.7B	1.7B	Instruct	12	1.762	0.917
Qwen3-4B	4B	Instruct	15	1.865	1.000
Qwen3-8B	8B	Instruct	21	1.851	1.000
Qwen3-1.7B-Base	1.7B	Base	14	1.745	0.917
Qwen3-4B-Base	4B	Base	18	1.818	1.000
Qwen3-8B-Base	8B	Base	21	1.871	1.000
Qwen3-4B-2507	4B	Instruct	21	1.885	1.000
Qwen2.5-7B	7B	Instruct	18	1.850	1.000
Llama-2-7B	7B	Base	9	1.797	1.000
OLMo-2-1B	1B	Base	9	1.767	1.000

Table 18: Multi-concept geometry: pairwise cosine similarities confirm approximate orthogonality ($|\cos| < 0.10$).

Model	Params	Type	Sent vs Truth	Sent vs Form	Truth vs Form	Mean $ \cos $
Qwen3-0.6B	0.6B	Instruct	0.073	-0.021	0.036	0.043
Qwen3-1.7B	1.7B	Instruct	0.094	0.076	0.095	0.089
Qwen3-4B	4B	Instruct	0.020	-0.006	0.067	0.031
Qwen3-8B	8B	Instruct	0.032	0.024	0.062	0.039
Qwen3-4B-2507	4B	Instruct	0.052	—	—	0.046
Qwen3-1.7B-Base	1.7B	Base	0.069	0.056	0.071	0.065
Qwen3-4B-Base	4B	Base	0.029	-0.010	0.052	0.030
Qwen3-8B-Base	8B	Base	0.028	—	—	0.040
Llama-2-7B	7B	Base	0.053	—	—	0.041
OLMo-2-1B	1B	Base	0.030	—	—	0.033

Table 19: Feature superposition measurement. Effective rank is stable (~ 13.5) despite $4\times$ variation in hidden dimension.

Model	Params	Type	Hidden Dim	Eff. Rank	Rank/Dim %	Sparsity
Qwen3-0.6B	0.6B	Instruct	1024	13.6	1.33%	0.611
Qwen3-1.7B	1.7B	Instruct	2048	13.5	0.66%	0.611
Qwen3-4B	4B	Instruct	2560	13.5	0.53%	0.615
Qwen3-8B	8B	Instruct	4096	13.3	0.33%	0.578
Qwen3-4B-2507	4B	Instruct	2560	13.5	0.53%	0.594
Qwen3-1.7B-Base	1.7B	Base	2048	13.3	0.65%	0.611
Qwen3-4B-Base	4B	Base	2560	13.5	0.53%	0.644
Qwen3-8B-Base	8B	Base	4096	13.3	0.32%	0.598
Llama-2-7B	7B	Base	4096	14.5	0.35%	0.710
OLMo-2-1B	1B	Base	2048	14.3	0.70%	0.663